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transported to the alveoli and released out as CO_2. Every 100mL of deoxygenated blood delivers 4mL of CO_2 to the alveoli for elimination.

Bohr effect and Haldane effect

Increase in pCO_2 and decrease in pH decrease the affinity of haemoglobin for oxygen and shifts the oxyhaemoglobin dissociation curve to the right and facilitates unloading of oxygen from hemoglobin in the tissue. This effect of pCO_2 and pH on the oxyhaemoglobin dissociation curve is called the Bohr small effect.

The Haldane effect, on the other hand describes how oxygen concentrations determines hemoglobin's affinity for carbon dioxide. The amount of carbon dioxide transported in blood is remarkably affected by the degree oxygenation of the blood. The lower the partial pressure of O_2 lower is the affinity of haemoglobin saturation with oxygen hence more CO_2 is carried in the blood. This phenomenon is called Haldane effect. This effects CO_2 exchanges in both the tissues and lungs.

In the lungs the process is reversed as the blood moves through the pulmonary capillaries, its pCO_2 declines from 45mm Hg to 40mm Hg. For this to occur carbondioxide is freed from HCO_3^- ions and Cl^- ions moves in to the plasma and reenters the RBC and binds with H^+ to form carbonic acid which dissociates in to CO_2 and water. This CO_2 diffuses along its partial gradient from the blood to the alveoli (Figure 6.8).

6.6 Regulation of Respiration

A specialised respiratory centre present in the medulla oblongata of the hind brain called respiratory rhythm centre

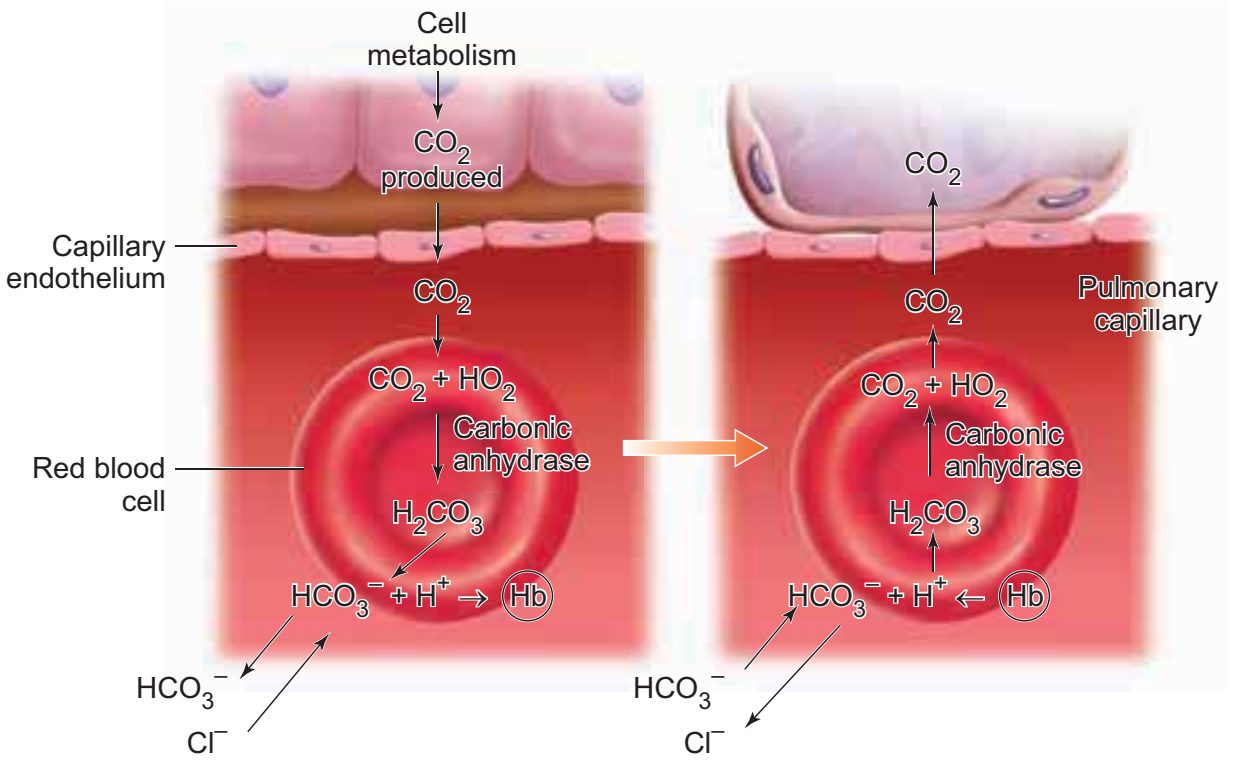


Figure 6.8 Chloride shift mechanism